

Fall 2026 Exploratory Data Analysis with R

Lecture 1: Introduction to R

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01

Course Logistics

01 Course Logistics

02 Getting Started

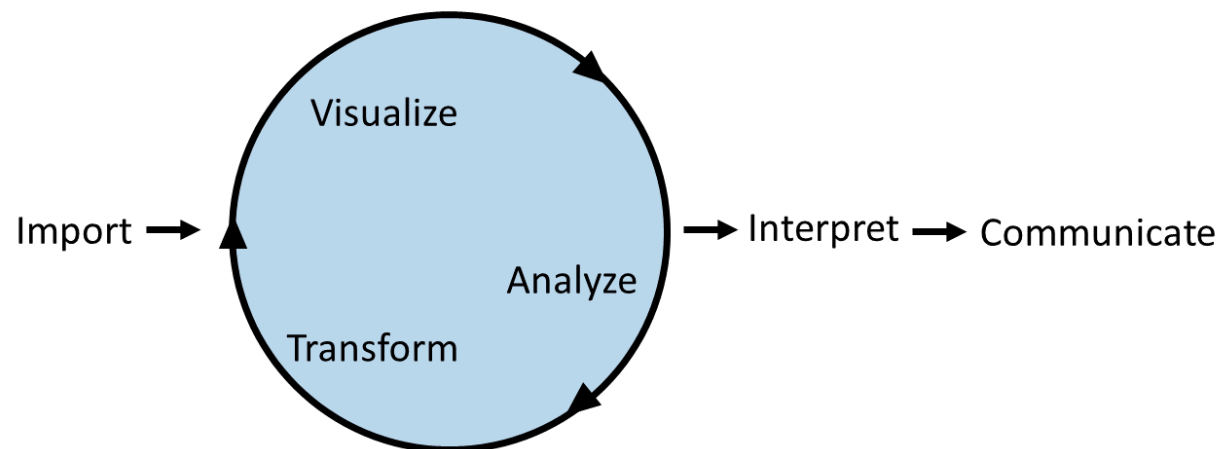
03 Import Data

04 Summarizing Data

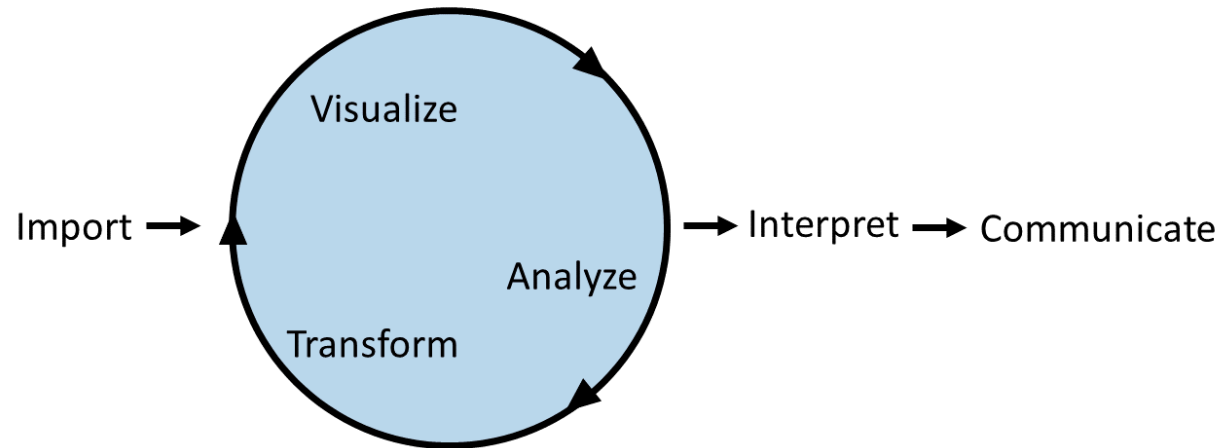
05 Visualizing Data

06 Recap

Big picture: Using data to understand our world



In this course, you will learn ...



- How to **think critically** about data and data analysis
- A programming language to **conduct data analysis**

How will you be evaluated in this course?

- Project: 65%
 - Status updates: 20%
 - First draft, presentation, final draft: 45%
- Homework assignments: 20%
- Quizzes: 10%
 - Graded check-ins at the beginning of every class
 - No exams!
- Communication, Timeliness, & Professionalism: 5%
 - A full grade reduction if you miss two scheduled meetings with your mentor without giving them advanced notice (usually 1 business day)
 - In-Class participation
 - Use respectful and professional language when communicating with course staff.
- **NOTE:** review the late submission policy

Academic Integrity

Per the [Spelman College Academic Integrity Policy](#) we expect that you will:

- Work collaboratively on assignments but that each person will do their own write-up of the solutions.
 - When working as a group: each person should think and work through each question.
 - Do not divide and conquer!
- Cite all your sources on your project; and that everything you write, even if it is just a summary of another paper, will be in your own words.

How can you get help in this course?

Learning a new programming language takes time. We provide as much support as possible.

- [Canvas](#), the primary source for information
 - Read the [“How to ask a question” handout](#) on Canvas
 - Post questions on the discussion boards
 - View Announcements
 - Download Lecture slides, Homework, Handouts
 - Review feedback when grades are posted
 - If you haven’t already, create an account and enroll via <https://canvas.instructure.com/courses/10046805>.
- Office hours (1-on-1 coaching and support)
- Send an email to:
- Talk to your economist advisor for questions about your hands-on project

Note: Atlanta Fed staff are only available during business hours

How can you help yourself in this course?

- Growth mindset, Iterative approach
 - *“Fail as quickly as possible, learn from it and move on” - Hadley Wickham*
- Built-in R documentation (more on this later)
- Course textbook: [R for Data Science](https://r4ds.had.co.nz) - r4ds.had.co.nz
- Reference [‘Cheat Sheets’](https://posit.co/resources/cheatsheets/) - https://posit.co/resources/cheatsheets/
- **Google**
 - “R [error message]”
 - “R [thing you want to do]”
 - “R [package or function name]”
- **ChatGPT**

Remember, data analysis is a process...

and there's actually no such thing as a “failure” or a “wrong answer”!

- Thinking critically about your code and your output is what gets you to the next step in the iterative process
- Remember, just because your code runs does NOT mean that it is correct
- An attempt will always get you at least partial credit
- An explanation of your thought process will get you more credit
- **Not attempting at all is the only reason you'll get a zero**

How do students describe this course?

- “It is both a course teaching the technical skill of R programming, as well as a practical lesson on how to conduct research.”
- “The course is really a great opportunity to practice data analysis from start to finish... Question, data, cleaning, analysis, presentation, write up. Other courses offer opportunities to write papers or do some segment of the process, but not the entire thing, and not with the regular practice.”
- “I really liked the attention and support that I got from the lecturers, the RA’s and other economists. It was intimidating at first but I got comfortable over time and I saw how my performance improved when I took advantage of all the help that was available to me.”

What you should expect for the course

- This course is extremely rewarding, but it will take effort.
- From the feedback we've received from students in past semesters, regarding the **effort, time** and **difficulty** of this course, we **require (much) more than that of the average class**.
- This information is not to discourage you, but to align your expectations with what we've gathered from students that have previously taken this class.

What advice would past students give...

1. **GO TO OFFICE HOURS**
2. Start the homework as soon as class ends
3. Don't wait until Thursday to do the homework
4. Go over the slides and homework before class and take notes, so that the time spent in class could be used to reinforce the information that was already read at home. It will also give you the opportunity to ask more questions in class

02

Getting Started

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Goals For Today

- Describe the steps of the data analysis process
- Understand how data are represented and stored
- Write and run R code using RStudio in your posit.cloud accounts
- Complete tasks in the data analysis process

What is RStudio?

RStudio is an Integrated Development Environment (IDE) for R that provides a graphical interface for the R software.

RStudio makes it easier to:

- write and run code
- preview plots and reports
- keep projects organized

What is Posit Cloud?

Posit Cloud is a cloud-based solution that hosts the RStudio IDE so it can be accessed from your browser.

Advantages:

- Key applications are already installed
- Standard hardware for everyone
- Helpful interaction features

This is our primary software in this course.

R, RStudio, and Posit Cloud – what is the difference?

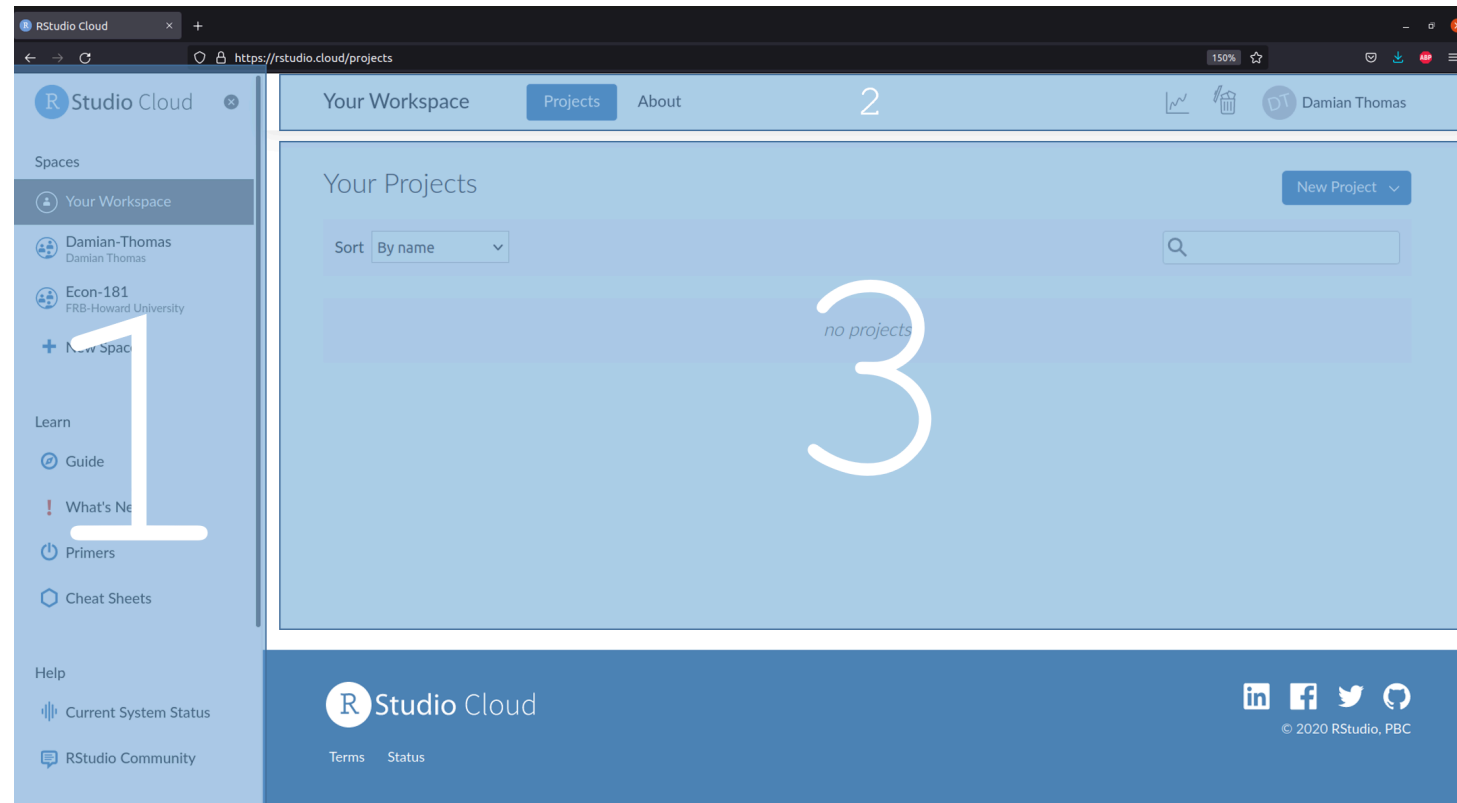
- First time users of R are often confused as to the difference between R and RStudio. RStudio is actually an add-on to R: it provides a user-friendly graphical interface for the R software.
- In this class, we use RStudio hosted in the Posit cloud which can be accessed directly from your browser.

Exercise 1: Create an Posit Cloud Account

You should have received an e-mail from posit.com with the Subject: Mark Jensen has invited you to access a space on Posit Cloud? It was sent to your university email account. There is a link in the e-mail that will take you to create a Posit Cloud account. Once you created your account and log-on you will be taken RStudio.

If you have already created a Posit Cloud account please log-on to it now. If you have not created an account please find this e-mail and do so now.

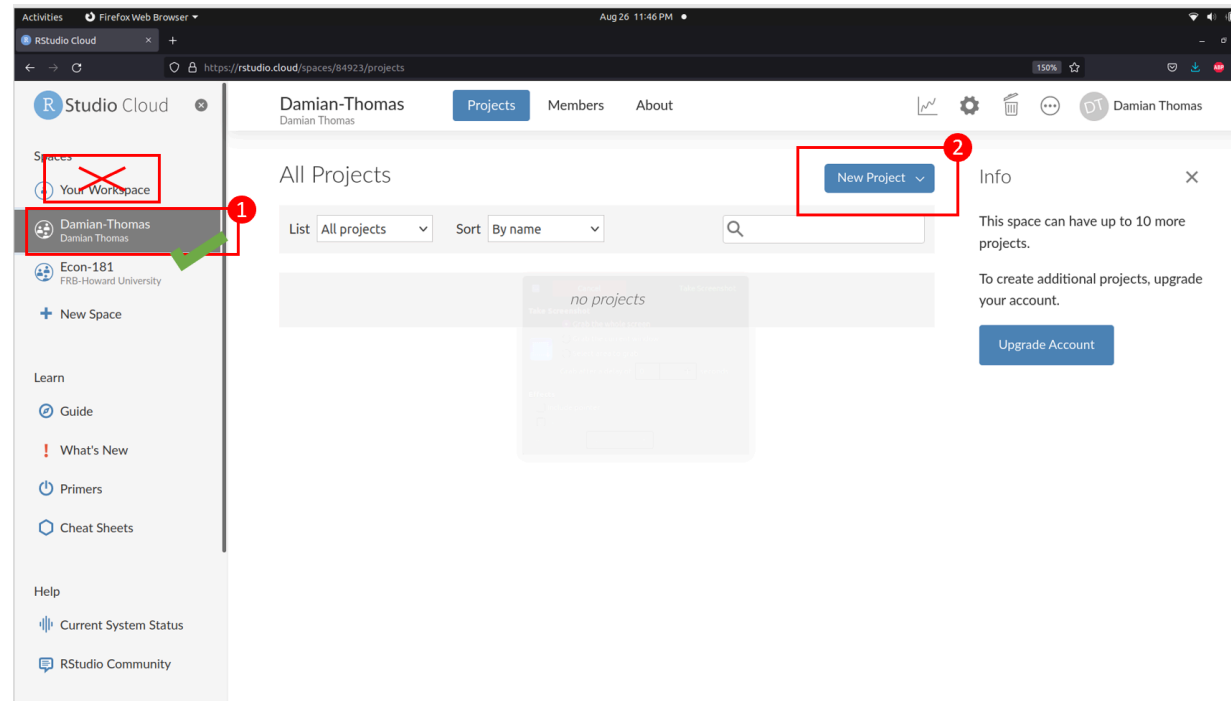
Posit Cloud's Interface



1: Navigation menu, 2: Rstudio header, 3: Project space

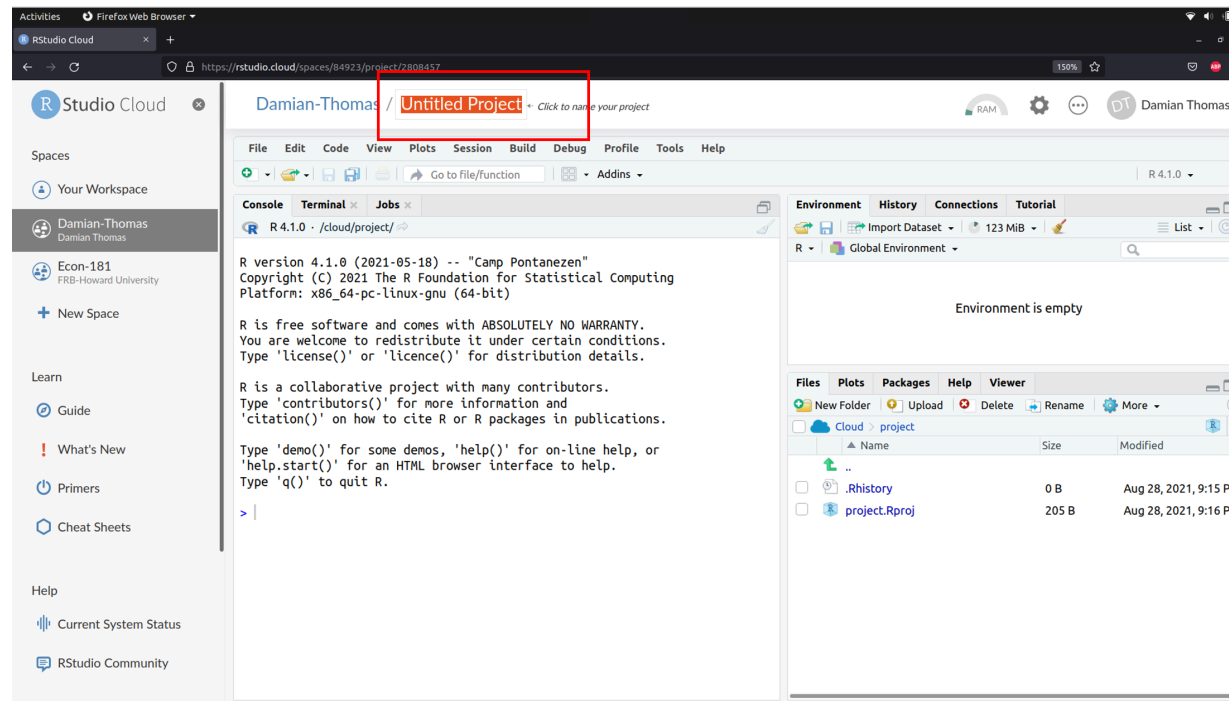
Create a Project

1. Select a Workspace in the navigation menu
2. Click on (First Name) 345 Workspace in the Spaces panel
3. Click on New Project located in the upper right-hand side

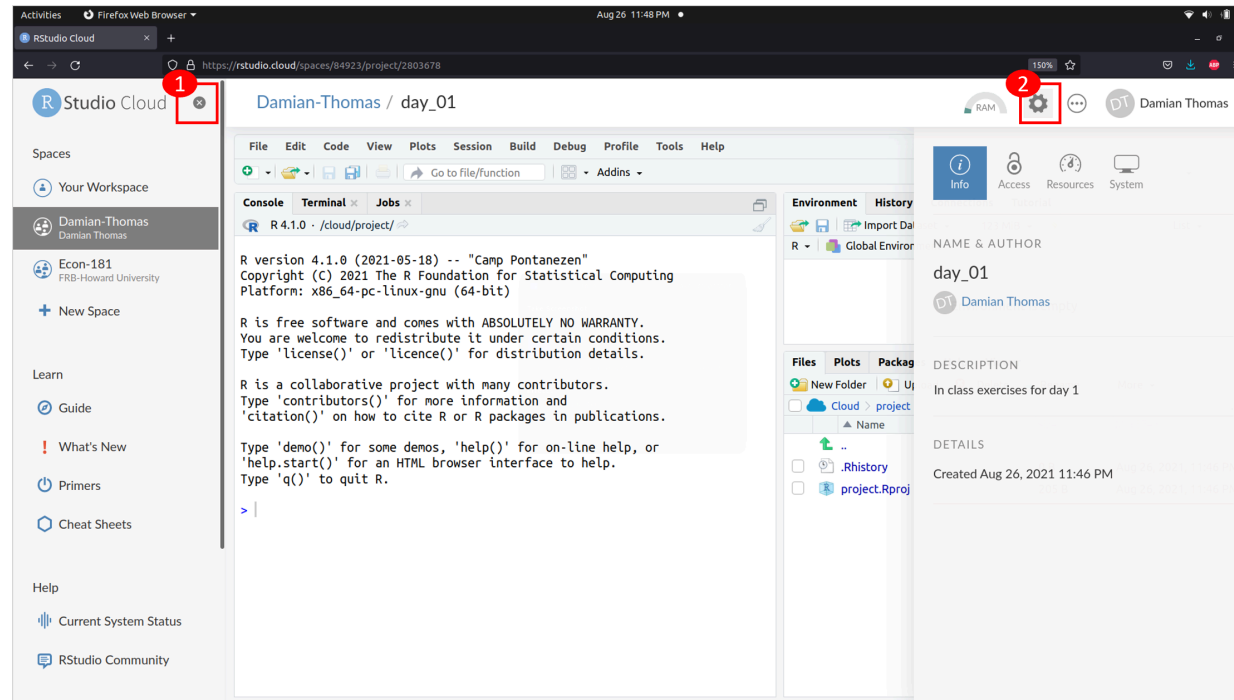


Rename the New Project

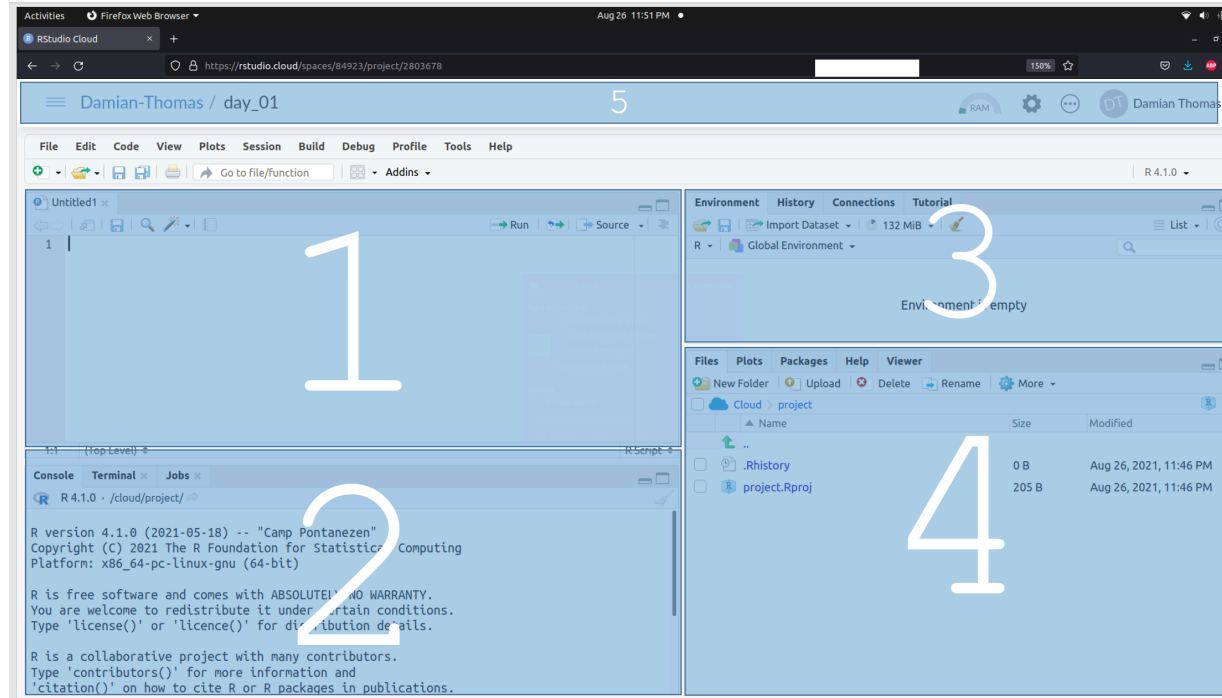
1. Click on the project title in the RStudio header
2. Type a new name



Hide/Unhide the sidebars



RStudio's Interface



1: Workspace, 2: Console, 3: Environment/History, 4: Files/Plots/Packages/Help

- Click on the **Help** tab, and search for “getwd”

Breakout Session 1

We will break here to review RStudio access and configuration. Please confirm that you have access to:

1. Your named Workspace. For instance, if I were enrolled in the class I would click on the Workspace entitled, “Mark 345 Workspace”.
2. Within this Workspace create a new Project and title it “Lecture 1 Project”.

03

Import Data

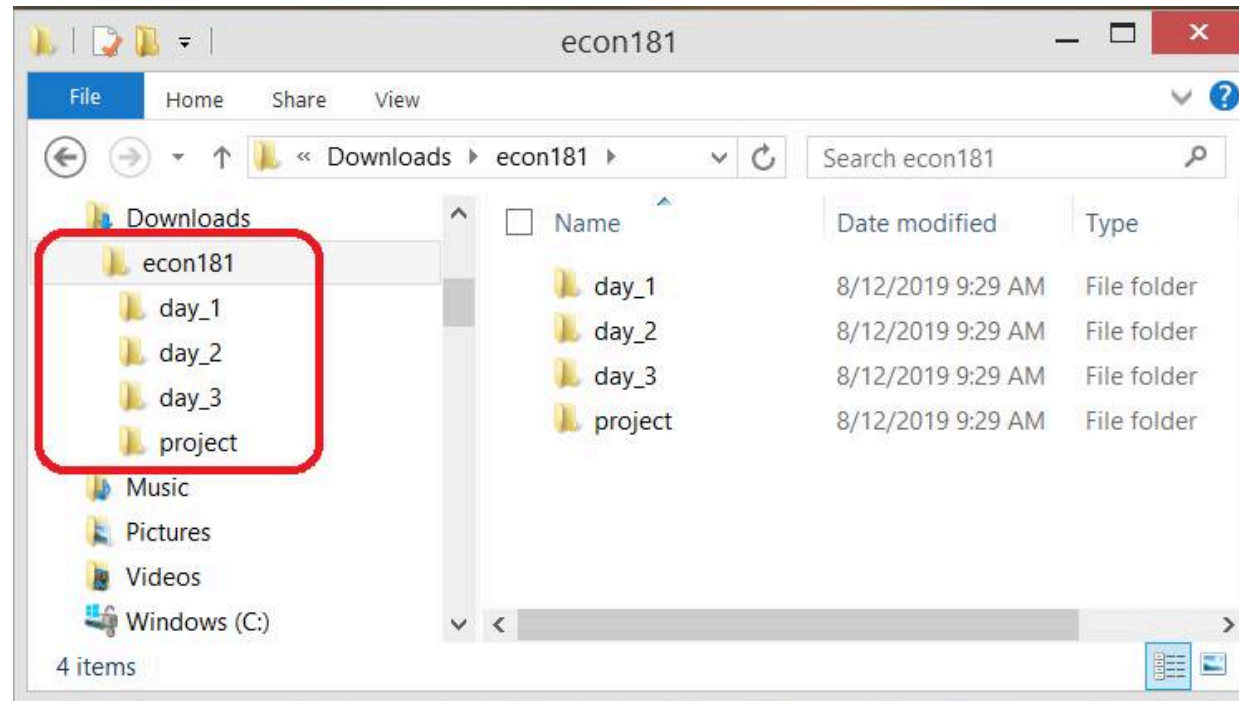
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Exercise 1: Import Data

1. We're going to find some data
2. Upload it to RStudio
3. Then tell R where it lives
4. Then read that data into R!

Directory Structure for Course Materials

- First, create a directory on your computer's hard drive
- You will need a organized place to store files on your computer



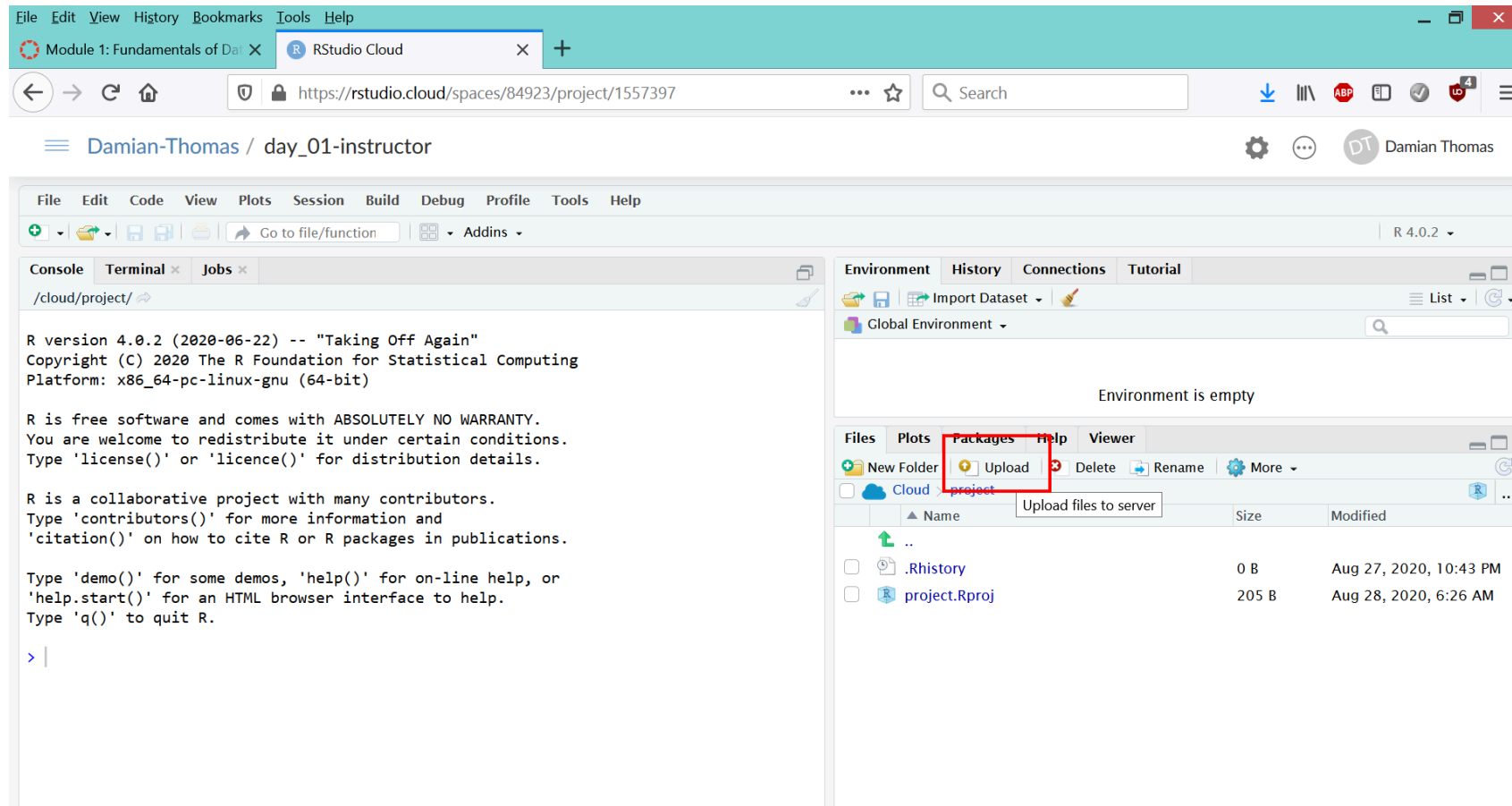
Download the data from Canvas

1. Go to [Canvas](#) and navigate to the **Module 1** page.
2. Click on [unemployment.csv](#) to download it.
3. Where should we save it?
 - in the directory we just created

What is a CSV file?

- A delimited text file that uses a comma to separate values.
- A file that stores tabular data in plain text.
 - Each line of the file is a single observation in the data.
 - Each observation consists of one or more variables, separated by commas.
- One of the simplest ways to save data and make it portable.
 - R, and all other common statistics/data software, can read in .csv files.

Upload the data to RStudio Cloud



Packages

A **package** is a collection of code and functions that other R users publish online. This course focuses on packages in the [tidyverse](#)

- Access the functions from a package

```
# Load a package in the current R session or program  
library(tidyverse)
```

- Download a new package, if needed

```
# Install a package on your computer  
install.packages("tidyverse")
```

Installing vs Loading

How do these commands differ?

	<code>install.packages()</code>	<code>library()</code>
What it does:	Get a new package	Enable use of new functions
When it is used:	One time per machine or project	In every session or program
Where to type it:	The Console prompt	The Script file

Comments in R Code

```
# Install a package on your computer  
install.packages("tidyverse")
```

```
# Load an installed package in the R session  
library(tidyverse)
```

- Text to the right of a `#` is considered a comment and ignored
- Other text is treated as code and executed

Import the data

```
# Import the raw data
unemp_df <- read_csv(file = "data/unemployment.csv")
```

What did we just do?

- We just read our data into R, creating a **data frame** called `unemp_df`

What is `read_csv()`?

- `read_csv()` is a function. Functions take **arguments** telling them what to **function** on!
- In this case, the **input** is the name of our data file (`"data/unemployment.csv"`)
- We **assigned** the **output** to the name `unemp_df` using the `<-` operator.
- What would happen if we misspelled the function name?

Leaving off and Picking up in R

1. Save your R script.

- Choose a name, `day_1.R`, for example.
- **WARNING: Do not to include spaces** in the file name. E.g.: '`coffee_and_cats.R`' is preferable to '`coffee and cats.R`'.

2. Close the script and return to your work space

Break

We will break here for 10 minutes.

When we come back, we'll have class introductions.

After the break: Class introductions

In alphabetical order (referencing the participants tab), everyone please introduce yourselves.

Students, please feel free to share:

- Your Spelman intro (name, year, major/degree, where you're from)
- What you are hoping to learn in this course

Staff introductions

Exercise 2: Resuming your work

Return to the project – it should look just like you left it.

```
library(tidyverse)
# read in data and name it unemp_df
unemp_df <- read_csv("data/unemployment.csv")
```

Protect your work. Save early and often. RStudio auto-save is not perfect!

Returning to a project

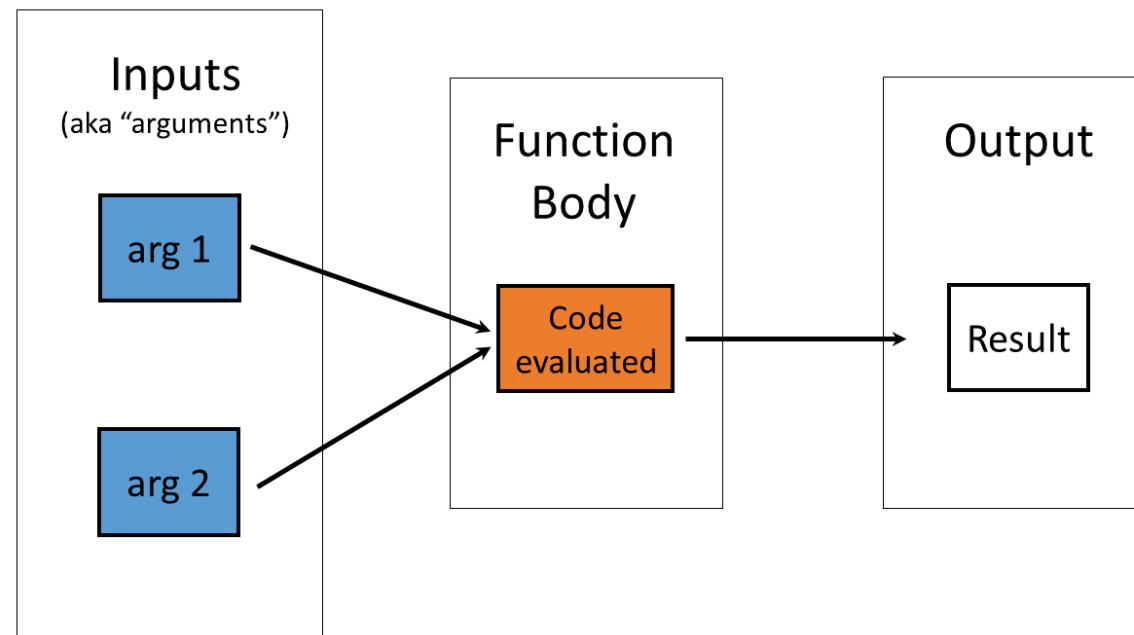
When returning to a project, make sure:

- The current working directory is correct
- Necessary packages are loaded
- The most recent version of the code was saved

These are small things to check for, but do so can help avoid unexpected result and lost time.

Functions are shortcuts for code

- Functions convert one or more inputs into a single output
- A function is essentially a named shortcut for stored code



Functions come from three sources:

- **base R** – functions included with R by default
- **user-defined** – functions we create ourselves
- **packages** – collection of code and functions that other R users publish online

Sum function example

```
# This function will print out the sum value of the numbers within the parenthesis  
sum(5, 7, 10)
```

```
[1] 22
```

Mini Recap

Some functions we have used

- `library()` - Loads a package into your session
- `setwd()` - Tells R where it should start looking for data files
- `read_csv()` - imports data from a csv file into a data frame your environment

Built-in Documentation

Access the help documents for any function using the `?` operator

Example: display help for the `head()` function

```
?head
```

Running the command provides

- An overview of what it does
- Outline usage with default options
- Detailed descriptions of each input argument and the output
- Examples of working code

Built-in Documentation

Now that we know how the head function works, let's use it to see it in action!

```
# head is very useful function  
head(unemp_df)
```

```
# A tibble: 6 × 2  
  Year Unrate  
  <dbl> <dbl>  
1  1948   3.75  
2  1949   6.05  
3  1950   5.21  
4  1951   3.28  
5  1952   3.02  
6  1953   2.92
```

What is in the data?

- Now that we've set our working directory and imported our data, what is in it?
- **What:** This is the unemployment rate in the United States
- **When:** 1948 - 2022
- How does R interpret the data? How will that affect what we can do with it?

What do our data look like?

- What if we want to look at the data?
- Type `View(unemp_df)` into your script (Note the capital V).

```
# glimpse is another useful function  
glimpse(unemp_df)
```

```
Rows: 75  
Columns: 2  
$ Year    <dbl> 1948, 1949, 1950, 1951, 1952, 1953, 1954, 1955, 1956, 1957, 195...  
$ Unrate  <dbl> 3.750000, 6.050000, 5.208333, 3.283333, 3.025000, 2.925000, 5.5...
```

Data Objects vs Functions

R has two main categories of objects:

- **Data object:** stores organized data for later use. Example object categories:
 - vector: a single value or an ordered collection of separate values
 - list: like a vector, but can also contain other objects
 - data frame: columns of data combined into a table (a list of vectors)
- **Function:** converts one or more *inputs* into a single output. Key characteristics:
 - body: instructions written in R code
 - arguments: receive inputs from outside the function
 - environment: temporary space where the function does its work

Note that it is common to hear the terms 'input' and 'argument' used interchangeably.

Data Objects vs Functions

Example data objects and functions as they typically appear in code

```
unemp_df <- read_csv(file = "data/unemployment.csv")
```



Data object
(*data frame*)



Function



Input
(*vector with one value*)

Data Types in R

Vector data types

Integer (int): whole numbers, including zero

-272, 0, 42

Numeric (num or dbl): decimal numbers

0.5, 3.17, 2.72

Character (chr): alphanumeric data (single or double quotes)

"keyboard", "USB", "1080p", 'Damian'

Logical (lgl): binary logic values

TRUE, FALSE

Date (date): calendar dates

"1867-03-02", "1913-12-23", "2000-01-01"

04

Summarizing Data

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Exercise 3: Summary Statistics

1. We're going to compute summary statistics
2. Then we're going to subset our data set
3. And then we'll combine these steps in a breakout session

What do our data look like, continued

Now that we have *imported* our data into R, what can we say about it?

- The `summary()` function computes several common summary statistics.

```
summary(unemp_df)
```

Year	Unrate
Min. :1948	Min. :2.925
1st Qu.:1966	1st Qu.:4.558
Median :1985	Median :5.567
Mean :1985	Mean :5.733
3rd Qu.:2004	3rd Qu.:6.846
Max. :2022	Max. :9.708

- What do we want to learn from our data?

Summarizing data meaningfully

What if we only want information from specific time periods?

- Suppose you only need the unemployment rate data **after** the year 2016.
- The `filter` function returns **only** the rows that meet specific conditions

```
filter(unemp_df, Year > 2016)
```

```
# A tibble: 6 × 2
  Year Unrate
  <dbl> <dbl>
1  2017   4.36
2  2018   3.89
3  2019   3.68
4  2020   8.09
```

```
5  2021    5.36
6  2022    3.67
```

- the first argument is the input data: `unemp_df`
- the second argument is the condition: `Year > 2016`

Breakout Session 2: Summarizing specific subsets of data

Breakout

- Use the filter and summary commands to return a summary of unemployment in the 1990s

Steps:

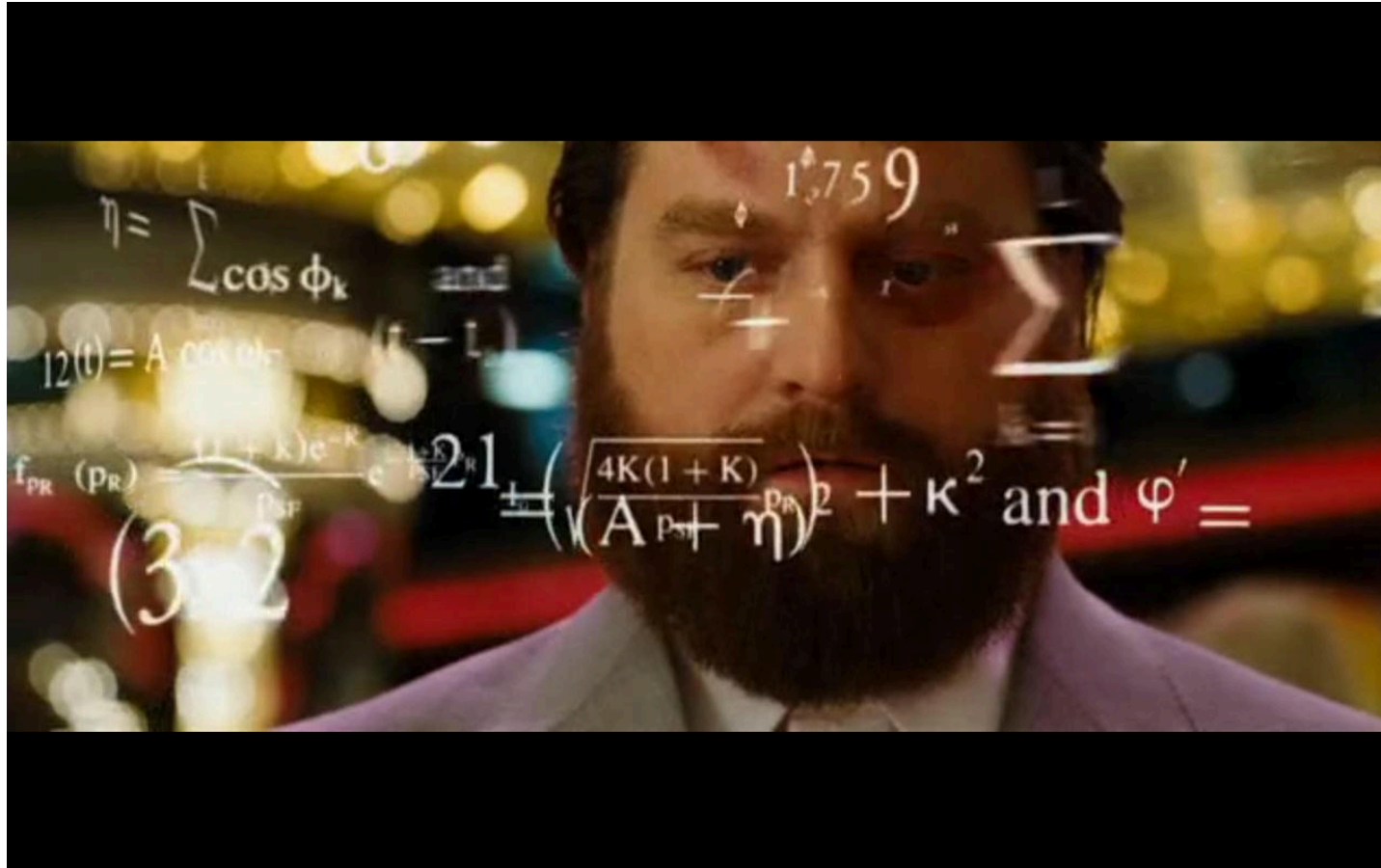
1. Identify our desired output (“What do we want”)
2. Find similar working methods (“What do we know that works”)
3. Identify what’s different (“How can we adapt what works”)
4. Run and repeat

Breakout Solution

```
summary(filter(unemp_df, Year >= 1990 & Year <= 1999))
```

Year	Unrate
Min. :1990	Min. :4.217
1st Qu.:1992	1st Qu.:5.058
Median :1994	Median :5.604
Mean :1994	Mean :5.763
3rd Qu.:1997	3rd Qu.:6.662
Max. :1999	Max. :7.492

Why summarize data?



Exercise 4: Analyze data

Compare two time periods

1. Compute mean unemployment for the time period 1980 - 1999
2. Compute mean unemployment for the time period 2000 - 2019
3. Calculate the difference. Discuss the results:
 - Economic significance?
 - Possible explanations?
 - Statistical significance? More about this later in the course.

Break

We'll stop here for another 10 minute break.

05

Visualizing Data

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Getting useful information

How else can we **effectively convey information**?

```
unemp_df <- read_csv("data/unemployment.csv")
```

Year	Average Unemployment
2015	5.3
2016	4.9
2017	4.4
2018	3.9
2019	3.7
2020	8.1
2021	5.4
2022	3.7

Data Visualization with `ggplot2`

- The `ggplot2` package provides a set of functions that enable all sorts of data visualizations.
- We will cover `ggplot2` in more detail in future classes.
- `ggplot2` is included in the `tidyverse` package, but to start using it you have to call it in a library function.

```
library(ggplot2)
```

Remind yourself what the data looks like

Using our `unemp_df` data:

```
library(ggplot2)
head(unemp_df)
```

```
# A tibble: 6 × 2
  Year Unrate
  <dbl> <dbl>
1  1948   3.75
2  1949   6.05
3  1950   5.21
4  1951   3.28
5  1952   3.02
6  1953   2.92
```

Plot the unemployment rate

We want to plot the unemployment rate as a number between 0 and 1

- Does this variable exist in the data already?
- How should it be computed?

Create a new variable

Use the `mutate` function:

```
# Add a new variable to unemp_df data frame  
unemp_df <- mutate(unemp_df, pct_unemp = Unrate/100)
```

Confirm the result

```
# Take a look at the changes  
head(unemp_df)
```

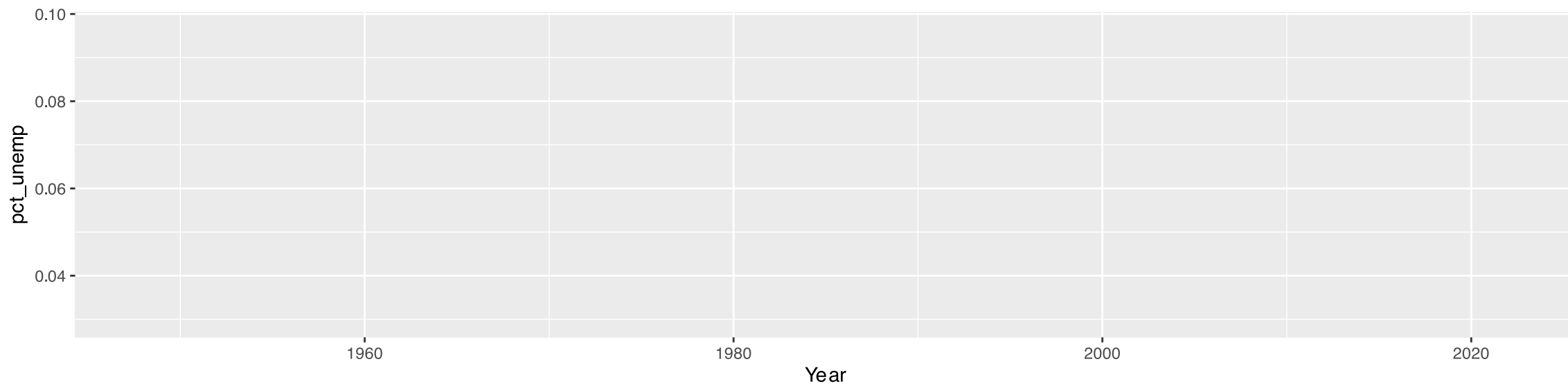
```
# A tibble: 6 × 3  
  Year Unrate pct_unemp  
  <dbl> <dbl>    <dbl>  
1  1948   3.75    0.0375  
2  1949   6.05    0.0605  
3  1950   5.21    0.0521  
4  1951   3.28    0.0328  
5  1952   3.02    0.0302  
6  1953   2.92    0.0292
```

Start with an empty plot

```
ggplot(data = unemp_df)
```

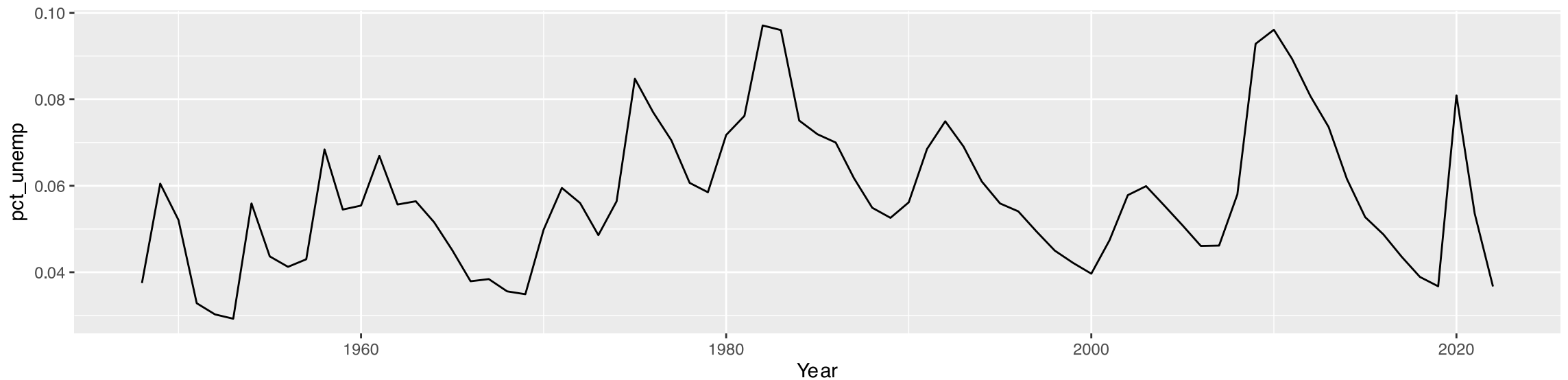
Define the x and y axis

```
ggplot(data = unemp_df,  
       mapping = aes(x = Year, y = pct_unemp))
```



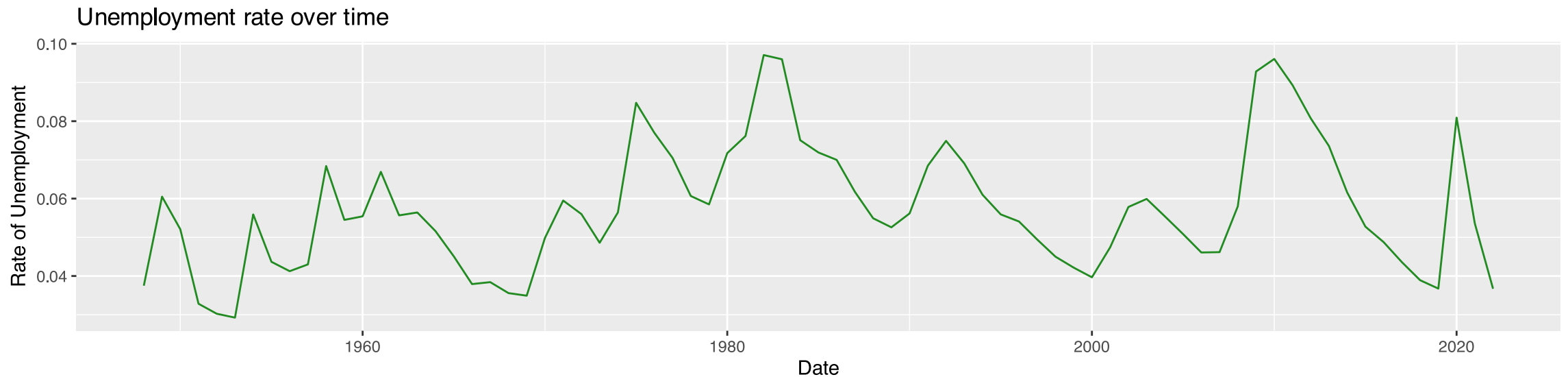
Add a line to the plot

```
ggplot(data = unemp_df,  
       mapping = aes(x = Year, y= pct_unemp)) +  
  geom_line()
```



Continue adding to the plot...

```
ggplot(data = unemp_df) +  
  geom_line(mapping = aes(x = Year, y = pct_unemp), color = "forestgreen") +  
  labs(x = "Date", y = "Rate of Unemployment", title = "Unemployment rate over time") +  
  theme(legend.title = element_blank())
```



W. E. B. Du Bois and the 1900 Paris Exposition

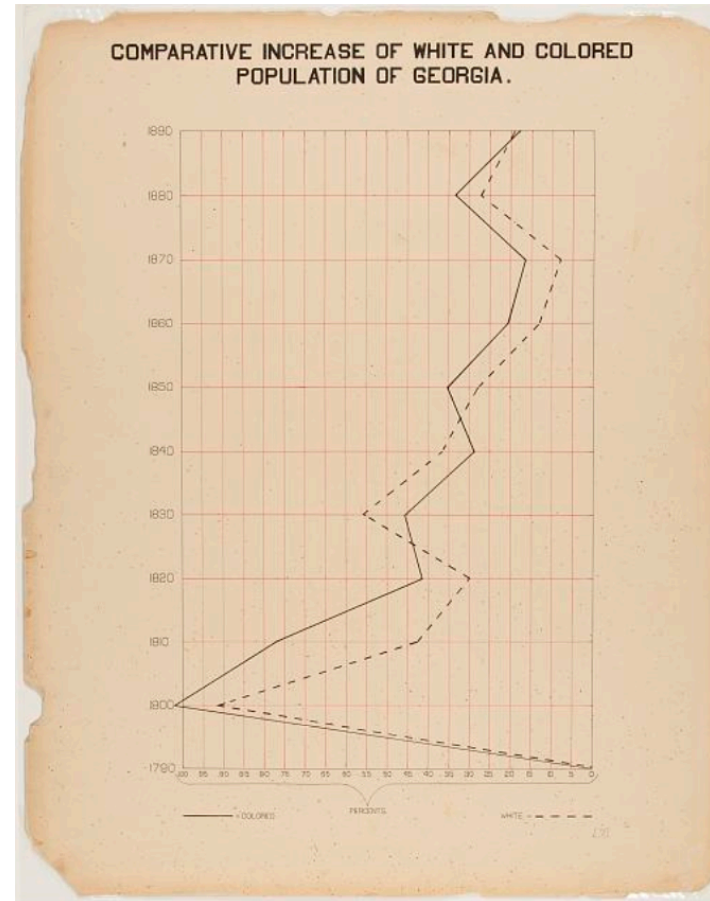
- The 1900 Paris Expo celebrated the achievements of the prior century
- Thomas Calloway¹ lobbied to become a Special Agent for the US delegation, spearheading an exhibit focused on African Americans
- W. E. B. Du Bois² organized and executed the exhibit which included more than 60 charts and statistical tables presenting demographic and economic data about African Americans. The exhibit also included several hundred photographs, scale models, and written works by African Americans.

Part of this week's homework will be based on these charts.

¹Calloway was an attorney and Howard Law School graduate

²Du Bois was a professor of History and Economics at Atlanta University which is now part of the Atlanta University Center along with Spelman

Images from the 1900 Paris Exposition



06

Recap

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Topics covered

- Downloading data
- Reading data into R
- Functions
- Data types
- Summarizing data
- Visualization

Functions Covered

```
head()  
glimpse()  
read_csv()  
setwd()  
getwd()  
install.packages()  
library()  
filter()  
mutate()  
ggplot()  
geom_line()  
summary()
```

Links

- [Canvas](#)
 - [Syllabus](#)
 - [Ask a TA](#) (discussion boards)
- [Posit Cloud](#)
- [Rstudio Cheatsheets](#)
 - [Data import](#) with the readr package
 - [Data Transformation](#) with the dplyr package